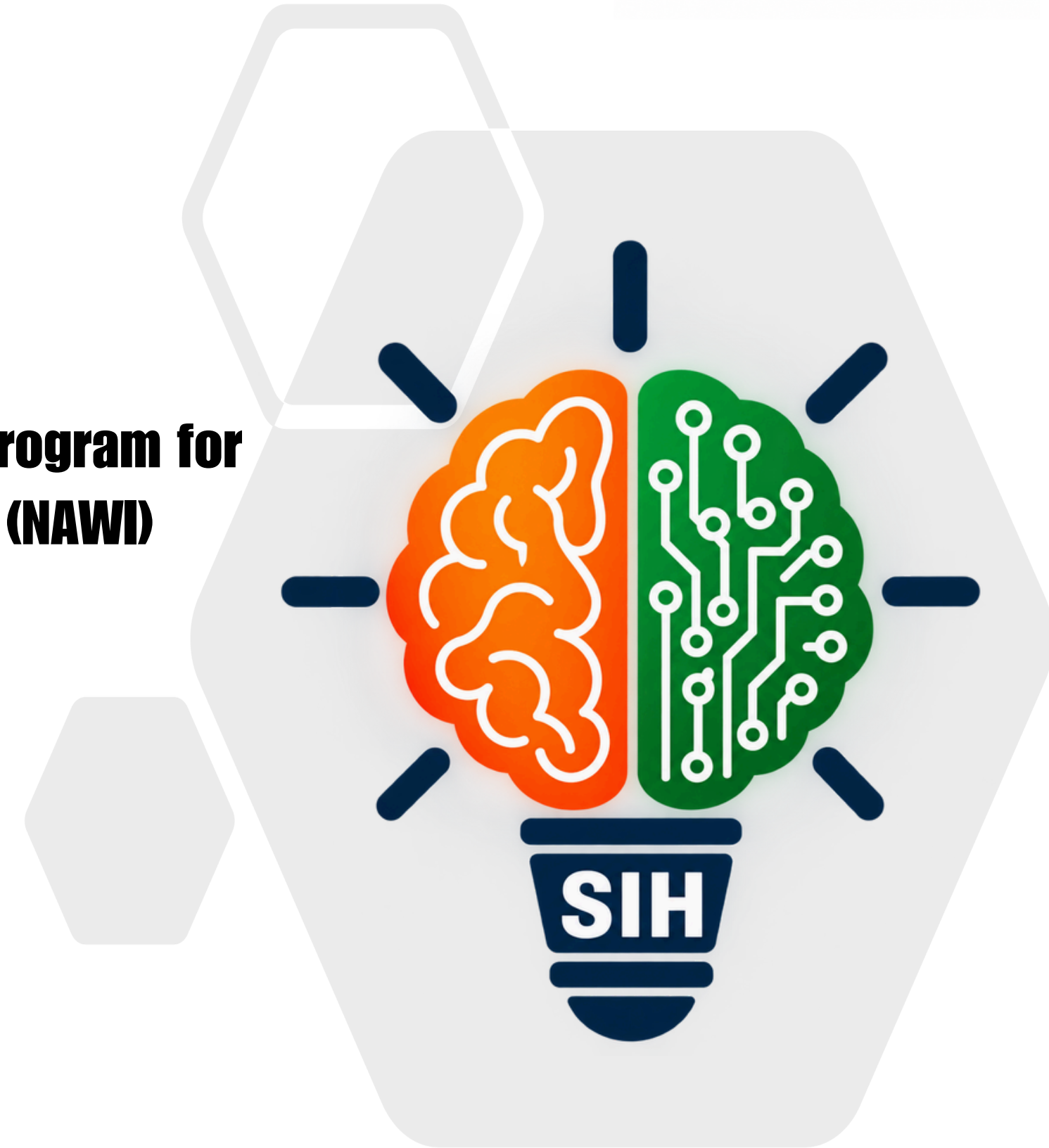


SMART INDIA HACKATHON 2026



- **ProblemStatement ID – 26035**
- **ProblemStatement Title- Development of a Software Program for generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R-76**
- **Theme- Miscellaneous**
- **PS Category- Software**
- **TeamID- 179282**
- **TeamName(Registered on portal) - _SpecCheck_**



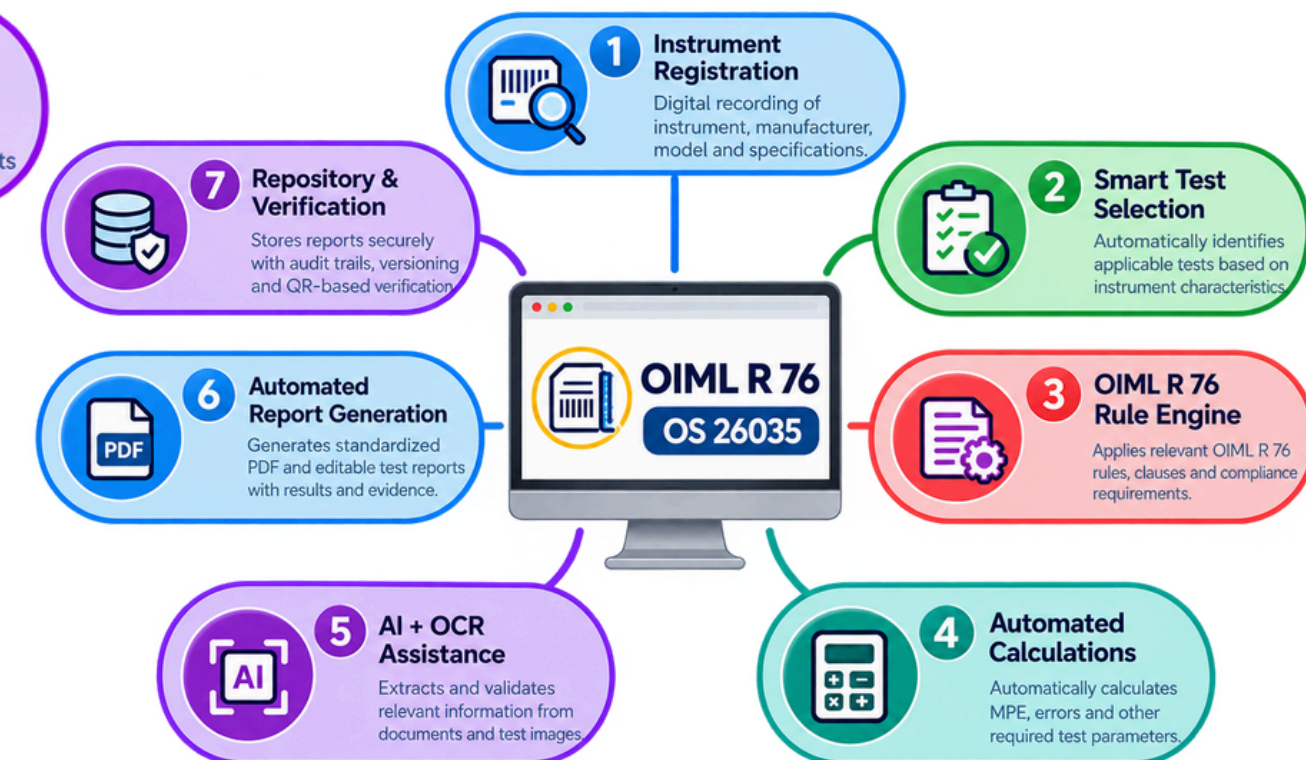
Proposed Solution

- **Digital NAWI Testing:** End-to-end digital workflow for instrument testing.
- **OIML R 76 Rule Engine:** Automated rule, clause and compliance checks.
- **Automated Calculations:** MPE, errors and other test parameters.
- **AI + OCR:** Extract and validate instrument/test data from documents and images.
- **Smart Test Selection:** Automatically identifies applicable tests.
- **Digital Evidence:** Store test images, certificates and supporting documents.
- **Automated Reports:** Generate PDF and editable OIML-based test reports.
- **Review & Approval:** Role-based verification and report finalization.
- **Repository & Dashboard:** Centralized reports, search, tracking and analytics.
- **Audit & QR Verification:** Complete traceability with version history and report verification.

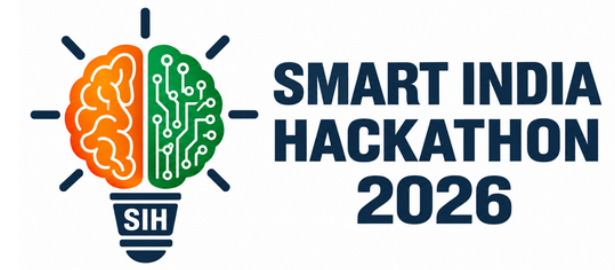


It Addresses the Problem By

- **Automates Manual Testing:** Converts the NAWI testing process into a structured digital workflow.
- **Reduces Errors:** Automates MPE calculations, validations and compliance checks.
- **Ensures Compliance:** Applies relevant OIML R 76 rules and clause-based requirements.
- **Saves Time:** AI/OCR-assisted data extraction and automated report generation.
- **Improves Traceability:** Centralized records, evidence, audit trails and QR-based report verification.



TECHNICAL APPROACH



Our Working Prototype : [click here](#)

Our GitHub repository : [click here](#)

Tech Stack

Website Frontend React.js TypeScript
Vite HTML5 CSS3

Backend REST APIs Python FastAPI

AI & CV OpenCV PaddleOCR Groq

Database Supabase PostgreSQL
Role-Based Access Control

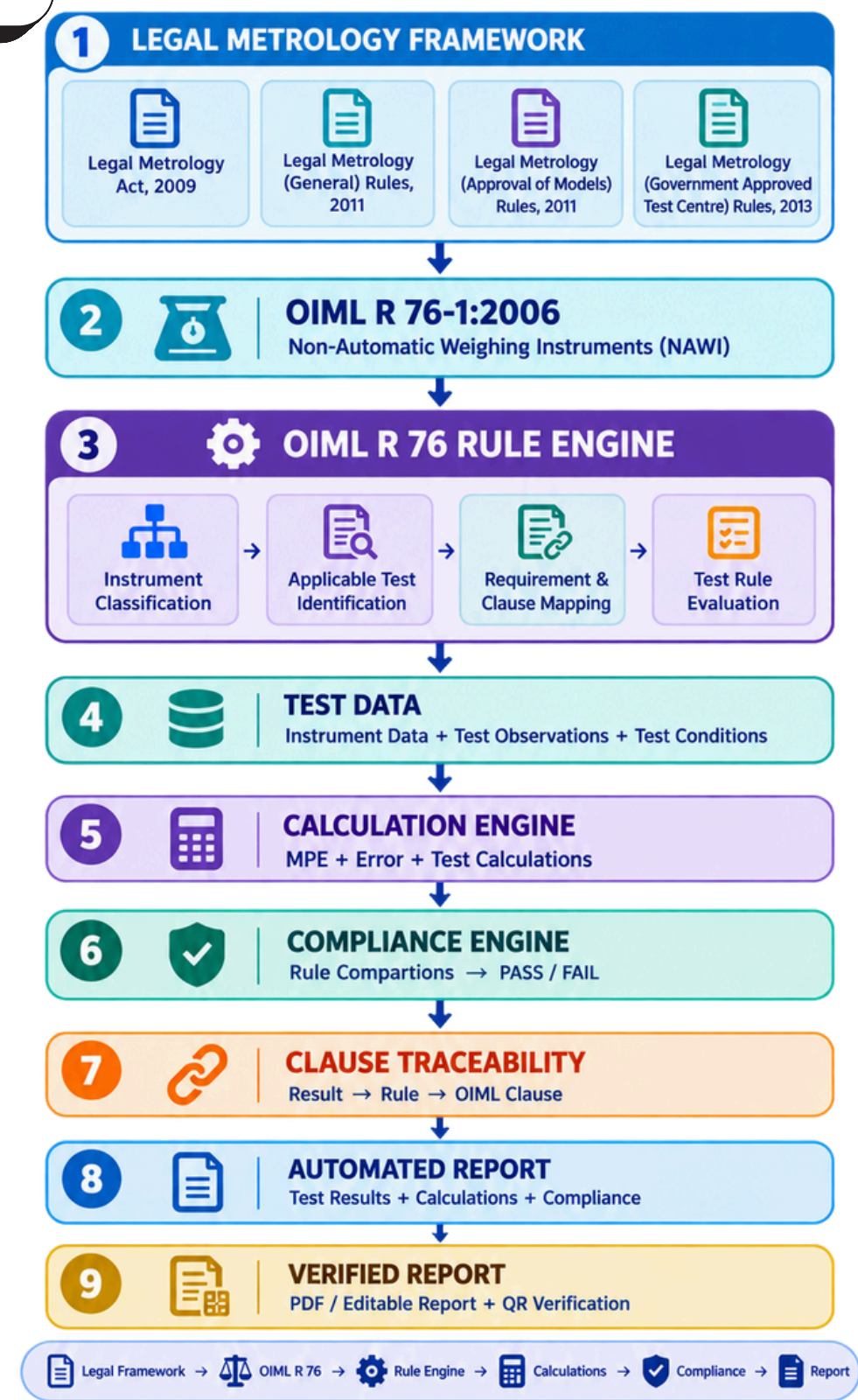
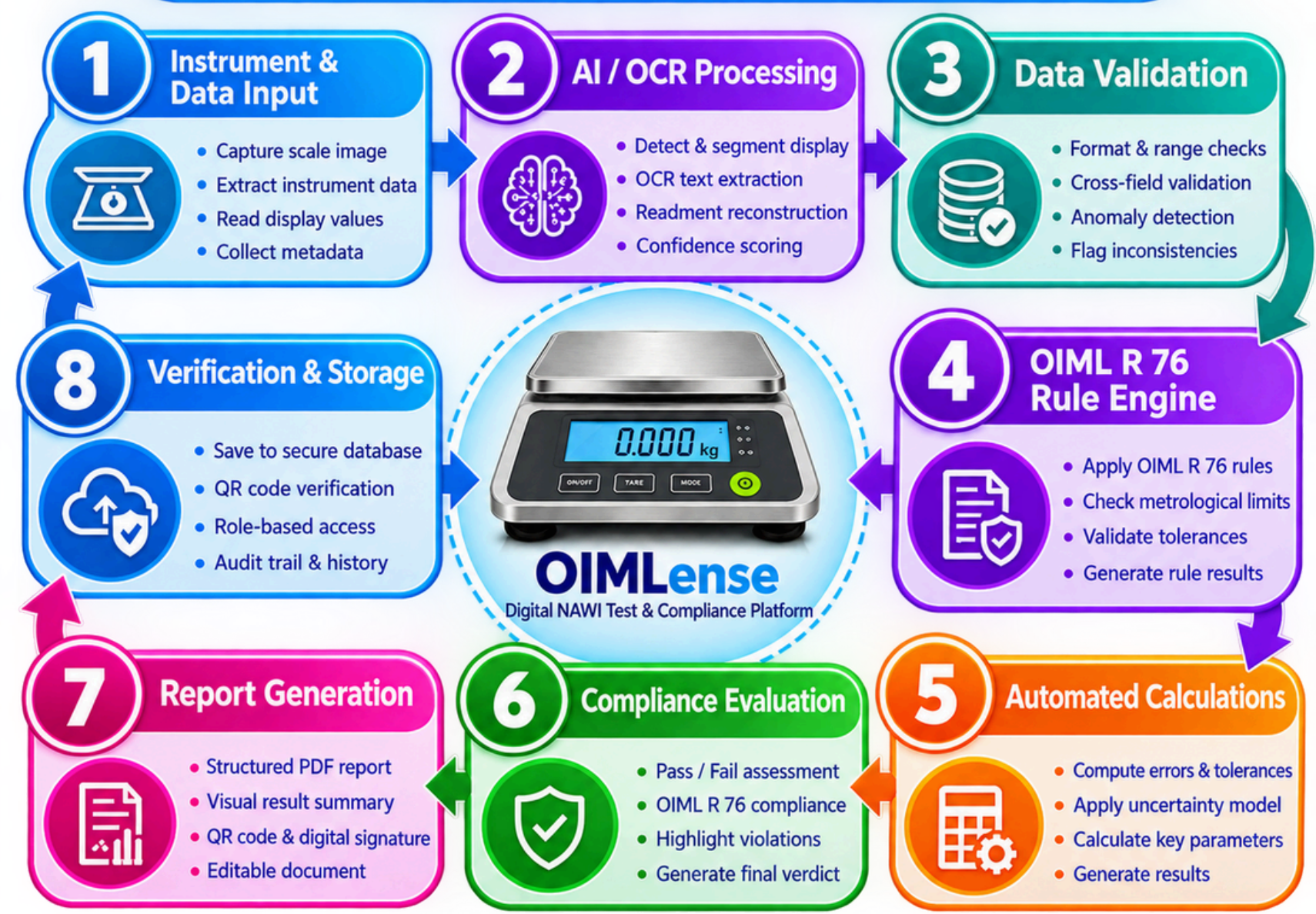
Reporting PDF generation
QR Code generation & verification
Editable document generation

Deploy & Dev Git GitHub Vercel
VS Code Cloud-based deployment

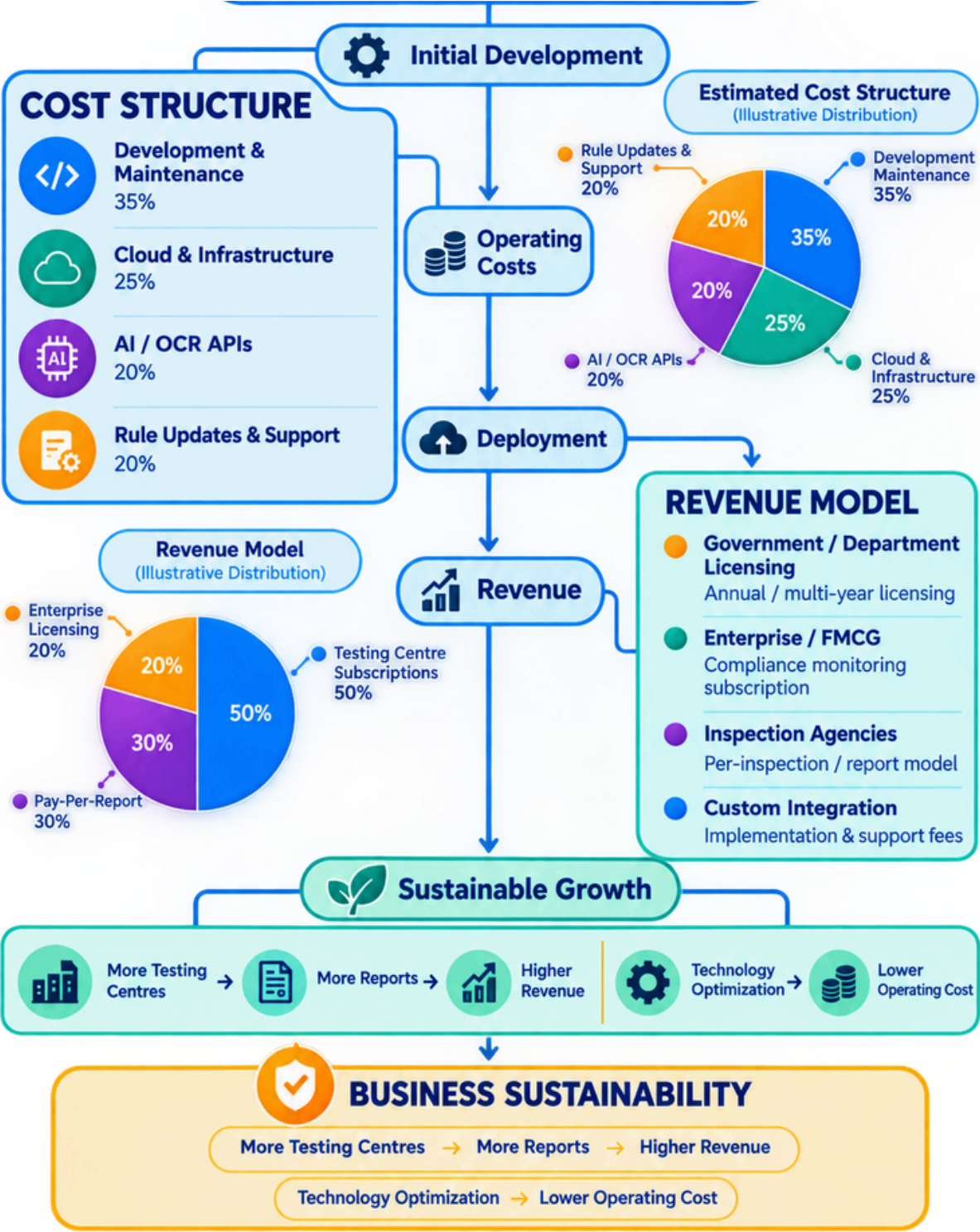
Modular, scalable tech stack adaptable to future frameworks

Technical Workflow – OIMLense

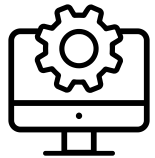
From Data to Trusted Compliance



Financial



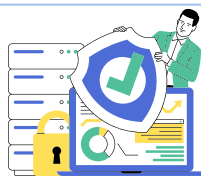
Feasibility



Uses proven technologies like Python, FastAPI, OCR and PostgreSQL.



OIML rules can be converted into digital rules.



Modules can be developed and updated independently.

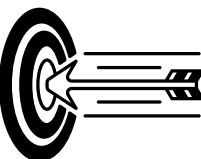


Can support multiple testing centres.

Viability



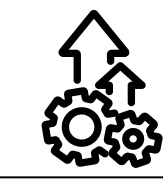
Automates testing and reporting tasks.



Reduces calculation and data-entry errors.



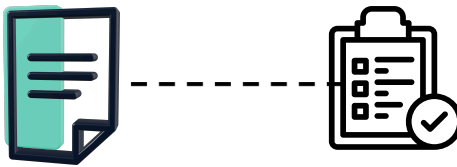
Speeds up testing and report generation.



Can support more instruments and standards.

Challenges And Mitigation

Complex Rules



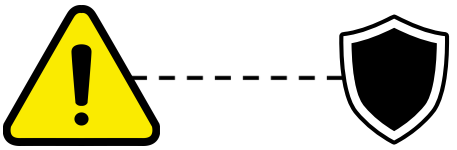
Complex OIML Requirements → **Clause-based modular rule engine**

Missing Data



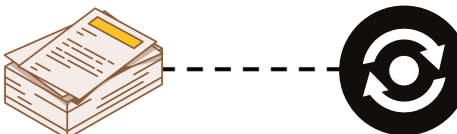
• Different Instruments Have Different Tests → **Instrument-specific test selection**

OCR Errors



• Manual Data May Contain Errors → **Automated validation & error checks**

Rule Updates



Regulations Can Change → **Version-controlled updateable rules**

IMPACT AND BENEFITS

SpecCheck



Converts traditional paper-based testing into a digital workflow.



Reduces the time required for testing and compliance evaluation.



Makes test results and compliance decisions easier to verify.



Promotes consistent application of testing requirements.



Provides secure and organized access to test reports.



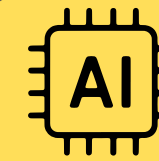
Enables adoption across multiple laboratories and testing centres.



IMPACT



BENIFITS



Automates repetitive testing and report preparation tasks.



Performs MPE, error and other calculations accurately.



Generates standardized reports with fewer manual errors.



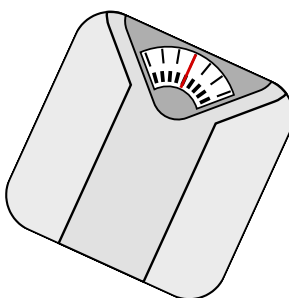
Automatically checks applicable OIML R 76 requirements.



Digitizes test data, documents and supporting evidence.



Maintains test history, versions and audit records.



Research Papers

- ◆ OIML R 76-1:2006 — Non-Automatic Weighing Instruments [\[Link\]](#)
- ◆ Legal Metrology Act, 2009 — Government of India [\[Link\]](#)
- ◆ Legal Metrology (General) Rules, 2011 [\[Link\]](#)
- ◆ Legal Metrology (Approval of Models) Rules, 2011 [\[Link\]](#)
- ◆ Legal Metrology (Government Approved Test Centre) Rules, 2013 [\[Link\]](#)
- ◆ Department of Consumer Affairs — Legal Metrology [\[Link\]](#)

Demo Video :
[click here](#)

GitHub repository :
[click here](#)

Working Prototype :
[click here](#)

Competition Comparison

	OIMLense	eMaap	e-Parimap	OIML R76 Remote
NAWI R-76 Focus	✓	✗	✗	✓
Automated MPE &	✓	✗	✗	✓
Automated Test	✓	⚠	⚠	✓
Compliance & Rule	✓	⚠	⚠	✓
Test Data & Traceability	✓	✓	✓	✓

Industry Outlook

- ₹54.3 Cr Legal Metrology investment
- 20 Units Testing equipment target
- 5% Verification growth target
- 3–4 Months Digital model approval
- 100+ OIML survey participants

Survey

- 85% Manual documentation is time-consuming
- 90% Prefer automated calculations
- 88% Need faster report generation
- 92% Consider digital traceability important